The replication code was developed using **Python 3.9** and **Stata 16.1**, and subsequently validated with Python 3.12.11 and Stata 17.0.107. However, compatibility issues may arise with other versions of Stata, such as Stata 14 or Stata 18, which may lack certain functions or employ different algorithms for the same function, potentially resulting in slightly different results. Therefore, it is strongly recommended to use the specific versions of Python and Stata (16) specified by the authors for replication purposes. The code was last run on a 2-core Intel-based laptop with macOS version 12.2.1 and 200GB of free space. The approximate runtime needed to reproduce the analyses on a standard 2025 desktop machine should be 10–60 minutes. The approximate storage space required should be less than 5GB.

List of tables and programs

Figure/Table	Program	Line Number	Output file
Table 1	T1T2/pca_revised.py		PCA_USA_197606_to_199001.xls
			PCA_USA_198912_to_202301.xls
			PCA_CHN_200404_to_202301.xls
Table 2	T1T2/pca_revised.py		PCA-UST ETF.xls
Table 3	T3new.do	164	T3_New_gmm_US_F1_1a.csv
		341	T3_New_gmm_US_F1_1b.csv
		496	T3_New_gmm_US_F1_1c.csv
		669	T3_New_gmm_US_F2_2a.csv
		844	T3_New_gmm_US_F2_2b.csv
Table 4	T4new.do	169	T4_New_gmm_US_F1_ZLB.csv
		367	T4_New_gmm_US_F2_ZLB.csv
Table 5	T5new.do	58	T5A.csv
		66	T5B.csv
		76	T5C.csv
		85	T5D.csv
		94	T5E.xlsx
Table 6	T6new.do	154	T6_New_gmm_China_F1_1a.csv
		310	T6_New_gmm_China_F1_1b.csv
		466	T6_New_gmm_China_F1_1c.csv
		621	T6_New_gmm_China_F2_2a.csv
		780	T6_New_gmm_China_F2_2b.csv
		936	T6_New_gmm_China_F2_2c.csv
Figure 1	Excel, Figure.xlsx		Excel, Figure.xlsx
Figure 2	Excel, Figure.xlsx		Excel, Figure.xlsx
Figure 3	Excel, Figure.xlsx		Excel, Figure.xlsx
Figure 4	Excel, Figure.xlsx		Excel, Figure.xlsx
Figure 5	Excel, Figure.xlsx		Excel, Figure.xlsx
Figure 6	Excel, Figure.xlsx		Excel, Figure.xlsx
Figure 7	Excel, Figure.xlsx		Excel, Figure.xlsx
Figure 8	Excel, Figure.xlsx		Excel, Figure.xlsx

The provided code reproduces all tables and figures in the paper.

All figures are located in the folder named "Figures," in the file "Figures.xlsx." If Excel displays the prompt “This workbook contains links to one or more external sources…”, please select “Don’t Update.”

Tables 1 and 2 can be produced using the folder T1T2 and the script **pca_revised.py**. This script produces summary statistics for US bonds (PCA_USA_197606_to_199001.xls, spreadsheet name: "10x10 corr PCA" [Table 1, Columns 1 to 3, Panel A] and PCA_USA_198912_to_202301.xls, spreadsheet name: "10x10 corr PCA" [Table 1, Columns 4 to 6, Panel A]), China bonds (PCA_CHN_200404_to_202301.xls, spreadsheet name: "10x10 corr PCA" [Table 1, Columns 7 to 9, Panel A]), and US ETFs (PCA-UST ETF.xls, spreadsheet name: "6x6 corr PCA" [Table 2, Panel A]) separately. The relevant column titles in these spreadsheets are: Factor 1, Factor 2, and Factor 3.

Panels B of **Table 1** and **Table 2** are located in the same corresponding Excel files, but under different spreadsheet names: "Performance measures". The relevant column titles in these spreadsheets are: Mean excess return in % [Mean], Excess return volatility in % [Vol], and Sharpe Ratio [SR].

The script **pca_revised.py** prompts the user to choose between US or China, Bond or ETF, and then to select the sample period. Note that to obtain the correct sample period, the user needs to extend the entry period by one month. For example, if the paper reports the sample period as 1990/01 to 2022/12, you should enter 1989-12 as the start date and 2023-01 as the end date. Therefore, the four inputs used to generate Table 1 and Table 2 are:
(1) USA, Bond, 197606, 199001;
(2) USA, Bond, 198912, 202301;
(3) CHN, Bond, 200404, 202301;
(4) USA, ETF, 200701, 202301.

These results are also summarized in the "T1T2 Reference Results" subfolder within the "T1T2" folder. For easier reference, the key results are highlighted in yellow in the Excel file, as the code produces additional results that are not included in the reported table.

Table 3 can be produced using the Stata do-file T3new. Running this do-file will generate five CSV files, each corresponding to one of the five columns in Table 3:

- Column 1: T3_New_gmm_US_F1_1a
- Column 2: T3_New_gmm_US_F1_1b
- Column 3: T3_New_gmm_US_F1_1c
- Column 4: T3_New_gmm_US_F2_2a
- Column 5: T3_New_gmm_US_F2_2b

The do-file contains five "cd" commands, where the user needs to specify their local directory paths.

Table 4 can be produced using the Stata do-file T4new. Running this do-file will generate two CSV files, each corresponding to two columns in Table 4:

- Columns 1 and 2: T4_New_gmm_US_F1_ZLB
- Columns 3 and 4: T4_New_gmm_US_F2_ZLB

In each CSV file, the results corresponding to the two table columns in the manuscript (representing one pair of results from the same estimation) are stacked into a single column in the CSV file. For example, in T4_New_gmm_US_F1_ZLB, variable names for Table 4's Column 1 appear as standard names (e.g., "Constant", "Level", "Slope"), while variable names for Column 2 have the "ZLB" prefix (e.g., "ZLBConstant", "ZLBLevel", "ZLBSlope"). These results are stacked in one column in the spreadsheet, but are presented as two separate columns in the manuscript's Table 4. The same structure applies for Columns 3 and 4 in T4_New_gmm_US_F2_ZLB.

The do-file contains two "cd" commands, where the user needs to specify their local directory paths.

Table 5 can be produced using the Stata do file T5new. The do file will produce five files: T5A.csv, T5B.csv, T5C.csv, T5D.csv, and T5E.xlsx, corresponding to the five rows of results in Table 5. In the manuscript, the coefficient of the Constant is reported first, while in the output, the coefficient of the Constant is reported last. Also, note that in the output files, the R^2 values are actual numbers, not percentages, whereas in the manuscript, the percentage values are reported. For Rows C and D, T5C.csv and T5D.csv separately report the OLS t-statistics and Newey-West t-statistics in Columns (1) and (2) of those csv files. This do file contains one "cd" command, where the user needs to specify their local directory.

Table 6 can be generated using the Stata do-file T6new. Running this do-file will produce six CSV files, each corresponding to one of the six columns in Table 6:

- Column 1: T6_New_gmm_China_F1_1a
- Column 2: T6_New_gmm_China_F1_1b
- Column 3: T6_New_gmm_China_F1_1c
- Column 4: T6_New_gmm_China_F2_2a
- Column 5: T6_New_gmm_China_F2_2b
- Column 6: T6_New_gmm_China_F2_2c

The do-file contains six "cd" commands, where the user should specify their local directory paths.

Dataset List

Data file	Source	Notes	Real Data Provided
T1T2/China_data.csv	WIND	As per terms of use	Yes
T1T2/F-F_Research_Data_Factors_CSV	Kenneth R. French Data Library	As per terms of use	Yes
T1T2/FRB_H15.csv	FRB	As per terms of use	Yes
T1T2/Treas ETF returns.xls	CRSP	As per terms of use	Yes
First regression data.xlsx	All listed	Extracted from the PCA analysis	Yes
MOVE.dta	Bloomberg	As per terms of use	Yes
T51.xlsx	All listed	Extracted from the PCA analysis	Yes
T52.dta	Bloomberg	As per terms of use	Yes
T53.dta	CBOE Archive	As per terms of use	Yes
T54.dta	CBOE	As per terms of use	Yes
tyVIX.dta	CBOE Archive	As per terms of use	Yes
VIXrevised.dta	CBOE	As per terms of use	Yes